

File permissions in Linux

Project description

Our Research Team needs to review the some files and directories in the 'projects' directory as the current permissions for some owner type are unnecessary, changing them will help us secure our system.

Check file and directory details

To check files and directory details we use : `ls -l`

Output:

```
researcher2@d7dab6743a08:~$ cd projects
researcher2@d7dab6743a08:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep 20 1
4:26 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 20 1
4:26 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 20 1
4:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 20 1
4:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 20 1
4:26 project t.txt
researcher2@d7dab6743a08:~/projects$
```

With this command we can see user and group and date and time.

Describe the permissions string

```
researcher2@d7dab6743a08:~/projects$ cd projects
researcher2@d7dab6743a08:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep 20 14:26 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Sep 20 14:26 project_k.txt
```

Like this string “drwx - - x - - -”, so this string has 10 characters.

The 1st character indicates whether we are implementing permissions on a file or a directory in this case character ‘d’ is for directory. For file you replace ‘d’ with ‘-’ hyphen.

The 2nd to 4th character shows ‘r’ for read, ‘w’ for write, ‘x’ for executable. 2nd to 4th characters are for the user/owner so in this case the user has read ,write, execute permissions over directory ‘drafts’.

The 5th to 7th characters indicate the group which has only ‘x’ execute permission and before and after it is a hyphen which means it does not have read and write permissions.

The 8th to 10th characters indicate the other user's owner type , all its characters are hyphens which means it does not have any permission for this directory.

Change file permissions

As from the 2nd section we can say that the ‘project_k.txt’ should not have write permissions for other users as it is only handled by the user and its group so we use this command: ‘chmod o-w project_k.txt’

-> chmod means change mode, o-w means take away write permission from others using minus sign.

Output:

```
researcher2@d7dab6743a08:~/projects$ chmod o-w project_k.txt
researcher2@d7dab6743a08:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep 20 14:26 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Sep 20 14:26 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Sep 20 14:26 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Sep 20 14:26 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Sep 20 14:26 project_t.txt
researcher2@d7dab6743a08:~/projects$
```

```
drwx--x--- 2 researcher2 research_team 4096 Sep 20 14:26 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Sep 20 14:26 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Sep 20 14:26 project_m.txt
```

The write permission of others have been successfully removed.

Change file permissions on a hidden file

We can see hidden files using this command: `ls -al`

Output:

```
researcher2@d7dab6743a08:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 20 14:26 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 20 15:22 ..
-rw--w---- 1 researcher2 research_team  46 Sep 20 14:26 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 20 14:26 drafts
```

There is a hidden file `‘.project_x.txt’` `‘.’` at the start indicates that the file is hidden.

So as the file is hidden or archived it means that any owner type should not be able to write or execute it but the user, group can read it so to remove it we again use this command like the above `chmod` command: `‘chmod u-w,g-w,g+r .project_x.txt’`

Output:

```
researcher2@d7dab6743a08:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@d7dab6743a08:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 20 14:26 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 20 15:22 ..
-r--r----- 1 researcher2 research_team  46 Sep 20 14:26 .project_x.txt
```

The only difference is we wrote dot before the file name and used `‘+’` sign which means adding a permission.

Change directory permissions

To change permissions on a directory we again use `chmod` command: `‘chmod g-x drafts’`

Output:

```
researcher2@d7dab6743a08:~/projects$ chmod g-x drafts
researcher2@d7dab6743a08:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Sep 20 14:26 drafts
```

Here we simply removed the execute permission from the group. As this directory only belong to the researcher2 aka the user.

Summary

Through linux commands we were able to see the permissions of files and directories as well as hidden files and were able to change the permissions so that only required owners can modify and see their work.